

Do AI Platform Agents Know What They Must Not Forget

Compaction Aware Intervention Before State Changing Operations

ARIN 5203 Course Project Proposal

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Long-running tool-using agents compact earlier interactions to control context length, latency, and cost. This can remove an approval boundary, prohibition, expiry, entity, amount, or recovery condition that determines whether the next tool call is safe. We ask whether a post-compaction evidence audit, applied immediately before a state-changing operation and under a fixed context budget, can choose Execute, Confirm, or Handoff more reliably than standard compaction methods. Our reading will connect stateful agent evaluation in τ -bench and ToolSandbox, pre-action review in SABER, evidence-preserving context management in U-Fold, and Governance Decay, which studies how compaction can erase visible policy constraints and lead to prohibited actions. We will build 20 base scenarios in a deterministic AI-platform simulator covering quota changes, workload preemption, resource transfer, model rollout, financial-style approvals, and delegated multi-agent actions. Each base produces three matched cases: complete evidence, one action-critical item removed, and equal-length non-critical noise removed, for 60 cases kept together by base during splitting. We will compare full history, a recent window, neutral and task-aware summaries, fixed constraint pinning, a U-Fold-inspired baseline, and our evidence-audit policy under the same raw-token budget. The audit permits Execute only when authority, scope, validity, negative constraints, and recovery evidence support the proposed operation; a single answerable gap triggers Confirm, while operator-only authority or unresolved policy conflict triggers Handoff. A deterministic state-transition oracle will score actual consequences. Quantitative evaluation will report harmful mutation, safe autonomous completion, critical-evidence retention, grounding precision, intervention accuracy, human-review burden, context reduction, latency, and token cost, with matched counterfactual tests and ablations. Qualitatively, the final report and demonstration will show a long trajectory, the evidence retained or lost during compaction, the resulting route, and the simulator state difference. Our team combines **Qingcheng Yu**'s experience in agent lifecycle and human-in-the-loop evaluation, **Siu Kwun Yeung**'s experience in enterprise multi-agent verification and audit, and **Qingyin Guo**'s experience in tool/API integration and agent orchestration. These backgrounds map directly to scenario design, evidence auditing, and simulator implementation. A small local pilot will validate feasibility, and synthetic results will not be presented as production guarantees.